

Jasper Sands

Palo Alto, CA | (650) 924-8429 | jaspersands02@gmail.com | js6908@columbia.edu
jaspersands.com | github.com/Jaspersands | linkedin.com/in/jaspersands

RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for qubit characterization and control.

EDUCATION

Columbia University, New York, NY Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advisor: Dr. Henry Yuen.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning.

Washington University in St. Louis, St. Louis, MO Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

TECHNICAL REPORTS

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* 2025.

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* Sep 2026 – May 2027

- Automating silicon spin qubit characterization under Dr. Nard Dumoulin Stuyck, targeting the manual tuning and interpretation of charge measurements that currently bottleneck semiconductor quantum hardware development.
- Evaluating the automated analysis against how an expert reads the same device by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks on live email traffic.

Desdr Open Insurance Toolkit, Columbia University · *Graduate Researcher* Sep – Dec 2025

- Built climate-risk and actuarial features from SMS survey data with Dr. Eugene Wu, estimating agricultural losses.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* May – Dec 2025

- Grounded generative protein modeling in molecular graph data under Dr. Venkat Venkatasubramanian.

Foster System Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation. Measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes, decoding syndromes client-side with exact MWPM and Union-Find. Finite-size scaling puts the threshold at 3.2% and 2.7%.

Q-Search: proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian hidden subgroup, dihedral coset, and linear code equivalence problems, with formal proof gates testing each proposed mechanism against 1,200+ classical baselines.

Noise-Aware Adiabatic Simulation: transverse-field Ising model · *Qiskit* IBM Heron hardware

- Digitized adiabatic ground-state preparation on Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps for second-order accuracy.

- Arithmetic coding driven by autoregressive language model predictions across five models on the Canterbury Corpus, reaching 5–16× average and 39× best-case compression against 2–4× for gzip and zstd.

TEACHING EXPERIENCE

- Introduction to Quantum Computing**, Columbia University · *Teaching Assistant* Jan – May 2026
- Parallel and Concurrent Programming**, Washington University in St. Louis · *Teaching Assistant* Jan – May 2025
- Reverse Engineering and Malware Analysis**, Washington University in St. Louis · *Teaching Assistant* Aug – Dec 2024

INDUSTRY EXPERIENCE

- Smack Technologies** · *AI Intern* May – Aug 2026
- Built physics and sensor simulation for the core pipelines, integrating the knowledge graph to expand corpus generation.
 - Benchmarked open-weight models head to head on speed, intelligence and concurrency, speeding corpus generation 64×.
 - Bounded simulator error to 0.55% against references chosen outside its own tests, in a report that regenerates when the physics changes.
- Highnote** · *Cybersecurity Engineer Intern* May – Jul 2024
- Designed and deployed the company-wide SIEM for a card-issuing fintech, ingesting over 1 TB/day.
- Mindtrip** · *Software Engineer Intern* May – Jul 2023
- Built internal admin and secure-query tooling through Retool, saving roughly 250 engineering hours a month.
- Bugcrowd** · *Associate Application Security Engineer* May – Jul 2020
- Triaged and validated vulnerability submissions from a global researcher community for enterprise clients.

AWARDS AND COMPETITIONS

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street and Classiq challenge.
- **Qualified**, NYU Abu Dhabi Quantum Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

TECHNICAL SKILLS

Quantum software: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution and benchmarking.

Quantum methods: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT and block encoding, linear combination of unitaries, quantum walks, VQE and QAOA, Trotter decomposition.

Error correction: Rotated and XZZX surface codes, stabilizer formalism, minimum-weight perfect matching and Union-Find decoding, threshold estimation, randomized benchmarking, tomography, noise modeling.

Programming and tools: Python, C++, Rust, C, MATLAB, JavaScript, Go, Java · PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy · Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025 – Jun 2026). Ran weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms lecture notes, each paired with simulation checks.
- **Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.
- **Team Captain**, WashU Gymnastics Club.
- **Student Organizer**, WashU Votes. Ran voter registration drives and lobbied at the Missouri State Capitol.